

Bridging econometrics and machine learning: Shaping a research agenda

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Problem statement

How can we apply machine learning the underlying philosophy of which is fundamentally instrumentalist in nature to domains involving the testing and parameterization of theoretical laws? This is the fundamental challenge that we face in applying machine learning to domains where the primary goal is not prediction but the experimental testing of hypotheses.



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Examples

- Physics
- Economics
- Biology
- etc.

Inverse modelling

- Forward problem
- Inverse problem

Forward problem

$$\text{Data} = F(p)$$

where p are the parameters or laws. Task is to predict the data. We can think of $F(p)$ as a machine that we wish to learn. Standard machine learning problems can be thought of as solving forward problems by learning the laws based on data and then assessing how well the learned model predicts the data.

Inverse problem

$$parameters(p) = F^{-1}(data)$$

we wish to recover the parameters or laws from the data. The data is the input not the output. the parameters or laws are the output.



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Economics and Econometrics

- structural estimation (a.k.a inverse problems) *effect* $\rightarrow$ *cause*, see Carrasco, M., Florens, J. P., Renault, E. (2007). Linear inverse problems in structural econometrics estimation based on spectral decomposition and regularization. Handbook of econometrics, 6, 5633-5751.
- reduced form estimation, concerned with identification (linear systems identification), causal inference: *cause* $\rightarrow$ *effect*, so this is a forward problem
- forecasting, concerned with prediction (forward problems)
- Machine learning (training: inverse problem, test/validation/inference: forward problem)

Some Bridges

Neural networks are a special case of sieve estimators. See

- Chen, X., & White, H. (1999). Improved rates and asymptotic normality for nonparametric neural network estimators. IEEE Transactions on Information Theory, 45(2), 682-691.
- Chen, X., & Shen, X. (1998). Sieve extremum estimates for weakly dependent data. Econometrica, 289-314.
- Farrell, M. H., Liang, T., & Misra, S. (2021). Deep neural networks for estimation and inference. Econometrica, 89(1), 181-213.
- Fabozzi, F. J., Fallahgoul, H., Franstianto, V., & Loeper, G. (2025). Asymptotic Properties of ReLU FFN Sieve Estimators. Studies in Nonlinear Dynamics & Econometrics, 29(6), 727-752.

Sieves

A sieve is a sequence of subsets of the parameter space indexed by sample size (Geman, 1985)
Because neural networks involve adding nodes as sample size increases in order to obtain a better fit, they meet the definition of a sieve.
Different sieve estimators employ different basis functions, e.g. activation functions.
Geman, S., & Hwang, C. R. (1982). Nonparametric maximum likelihood estimation by the method of sieves. The annals of Statistics, 401-414.



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Sieves in structural estimation

- Forneron, J. J. (2023). A Sieve-SMM Estimator for Dynamic Models. Econometrica, 91(3), 943-977.
- Menzel, K. (2021). Structural sieves. arXiv preprint arXiv:2112.01377.

Shapes

Many substantive fields involve shape restrictions on empirical models that are derived from theory. essentially, this means we wish to impose shape restrictions on the target function that we are tying to learn. given that the neural network can learn any functional form, we want it to learn functions that at least fall into the right class based on domain knowledge.
So how do we do that with a neural network?



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Shapes: some literature

The literature on shape constraints spans both econometrics and machine learning.

- Aubin-Frankowski, P.-C. Szab´o, Z. (2020), 'Hard shape-constrained kernel machines', Advances in Neural Information Processing Systems 33, 384–395.
- Aubin-Frankowski, P.-C. Szab´o, Z. (2022), 'Handling hard affine SDP shape constraints in RKHSs', Journal of Machine Learning Research 23(297), 1–54
- Chetverikov, D., Santos, A. Shaikh, A. M. (2018), 'The econometrics of shape restrictions', Annual Review of Economics 10(1), 31–63.
- Gupta, A., Marla, L., Sun, R., Shukla, N. Kolbeinsson, A. (2021), Pender: Incorporating shape constraints via penalized derivatives, in 'Proceedings of the AAAI Conference on Artificial Intelligence', Vol. 35, pp. 11536–11544.

Shapes, continued

- Gupta, M., Bahri, D., Cotter, A. Canini, K. (2018), 'Diminishing returns shape constraints for interpretability and regularization', Advances in neural information processing systems 31.
- Gupta, M., Louidor, E., Mangylov, O., Morioka, N., Narayan, T. Zhao, S. (2020), Multidimensional shape constraints, in 'International Conference on Machine Learning', PMLR, pp. 3918–3928.
- Johnson, A. L. Jiang, D. R. (2018), 'Shape constraints in economics and operations research', Statistical Science 33(4), 527–546.

Applications

- Demand system estimation, e.g. Almost Ideal Demand Systems estimated non-parametrically using machine learning techniques. Hopefully with better predictive power.
- Augment this with law of demand constraints (Milleron-Mityushin Polterovich theorem) - a monotonicty restriction.



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Thanks for listening!

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